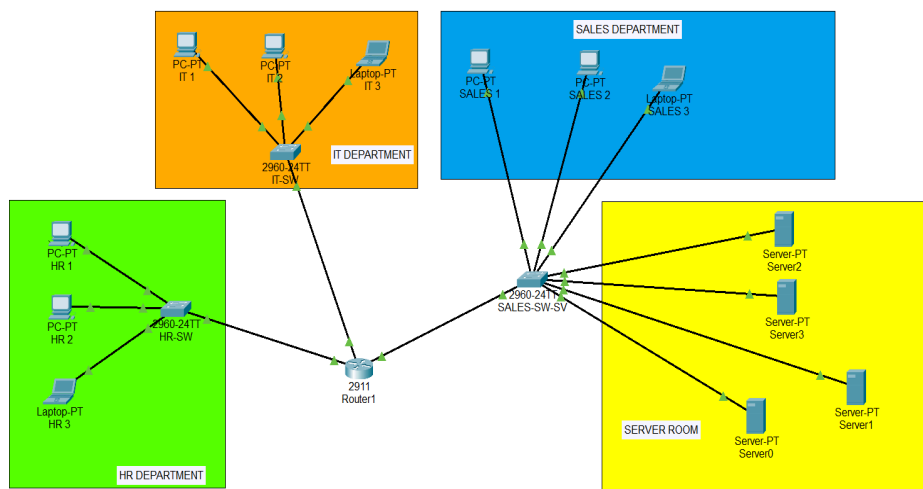


Cisco Packet Tracer Enterprise Network Setup with VLANs, DHCP, and Inter-VLAN Routing

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Objective

Design and implement an enterprise-level network using Cisco Packet Tracer with proper network segmentation using VLANS and DHCP for assigning dynamic IPs and router-based inter-VLAN communication.

Network Design Overview

DEPARTMENTS	NETWORK ADDRESS
HR	192.168.10.0/24 (VLAN 10)
IT	192.168.20.0/24 (VLAN 20)
SALES	192.168.30.0/24 (VLAN 30)
SERVERS	192.168.99.0/24 (VLAN 99)

Devices

- **Router: ISR4331**(handles inter-VLAN routing and DHCP)
- **Switches:**
 - 1 **Cisco 2960 switch** for HR(connected to router interface **GigabitEthernet0/0**)
 - 1 **Cisco 2960 switch** for IT(connected to router interface **GigabitEthernet0/1**)
 - 1 **Cisco 2960 switch** for SALES and SERVERS(connected to router interface **GigabitEthernet0/2**)

- **End Devices per Department:** 2 PCs + 1 Laptop.
- **Server room:** Dedicated to servers(using the same switch as SALES)

Topology Notes

- The switch for HR and IT uses access ports to connect to the router, which will handle a single VLAN from both departments.
- SALES and SERVERS share a single switch, using a trunk port to the router with subinterfaces(**Gig0/2.30** for SALES, **Gig0/2.99** for SERVERS).
- Each department is segmented into its own VLAN with DHCP services provided by the router.

Step-by-step Implementation

Step 1: Physical Topology Setup

Connect Devices:

- HR Switch: I connected 2 PCs and 1 Laptop to **FastEthernet0/1-0/3** and the switch to the router's **GigabitEthernet0/0**.
- IT Switch: I connected 2 PCs and 1 Laptop to **FastEthernet0/1-0/3** and the switch to the router's **GigabitEthernet0/1**.
- SALES/SERVERS Switch: I connected 2 PCs and 1 Laptop(SALES) to **FastEthernet0/1-0/3**, and server(s) to **FastEthernet0/4-0/6**. I connected the switch to the router's **GigabitEthernet0/2**.

Cabling:

- Used copper straight-through cables for PC/laptop/server-to-switch connections.
- Used copper straight-through cables for switch-to-router connections.

Step 2: VLAN Configuration on switches

Create VLANS:

- On HR Switch:

Plaintext

```
vlan 10  
name Vlan_HR
```

- On IT Switch:

Plaintext

```
vlan 20  
name Vlan_IT
```

- On SALES/SERVERS Switch:

Plaintext

```
vlan 30
name Vlan_SALES

vlan 99
name Vlan_SERVERS
```

Configure Access Ports (for end devices) :

- On HR Switch:

Plaintext

```
Interface range FastEthernet0/1-3
Switchport mode access
Switchport access vlan 10
No shutdown
exit
```

- On IT Switch:

Plaintext

```
Interface range FastEthernet0/1-3
Switchport mode access
Switchport access vlan 20
No shutdown
exit
```

- On SALES/SERVERS Switch:

Plaintext

```
Interface range FastEthernet0/1-3
Switchport mode access
Switchport access vlan 30
No shutdown
Exit

Interface range FastEthernet0/4-6
Switchport mode access
Switchport access vlan 99
```

```
No shutdown
exit
```

Configure Switch-to-Router Ports:

- On HR Switch: (access port for VLAN 10)

Plaintext

```
Interface GigabitEthernet0/1
Switchport mode access
Switchport access vlan 10
No shutdown
Exit
```

- On IT Switch: (access port for VLAN 20)

Plaintext

```
Interface GigabitEthernet0/1
Switchport mode access
Switchport access vlan 20
No shutdown
Exit
```

- On SALES/SERVERS Switch: (trunk port for VLANs 30 and 99):

Plaintext

```
Interface GigabitEthernet0/1
Switchport mode trunk
Switchport trunk allowed vlan 30,99
No shutdown
Exit
```

Step 3: Router Configuration :

- Configure Router Interfaces:

Plaintext

```
hostname Router
!
interface GigabitEthernet0/0/0
ip address 192.168.10.1 255.255.255.0
no shutdown
!
interface GigabitEthernet0/0/1
ip address 192.168.20.1 255.255.255.0
no shutdown
!
interface GigabitEthernet0/0/2.30
encapsulation dot1Q 30
ip address 192.168.30.1 255.255.255.0
no shutdown
!
interface GigabitEthernet0/0/2.99
encapsulation dot1Q 99
ip address 192.168.99.1 255.255.255.0
no shutdown
```

- Enable IP Routing (for inter-VLAN Communication):

```
ip routing
```

Step 4: DHCP Configuration on Router

- Exclude Reserved IP Addresses:

```
ip dhcp excluded-address 192.168.10.1 192.168.10.10
ip dhcp excluded-address 192.168.20.1 192.168.20.10
ip dhcp excluded-address 192.168.30.1 192.168.30.10
ip dhcp excluded-address 192.168.99.1 192.168.99.10
```

- Configure DHCP Pools:

```
ip dhcp pool HR
network 192.168.10.0 255.255.255.0
default-router 192.168.10.1
dns-server 8.8.8.8
!
ip dhcp pool IT
network 192.168.20.0 255.255.255.0
default-router 192.168.20.1
dns-server 8.8.8.8
!
ip dhcp pool SALES
network 192.168.30.0 255.255.255.0
default-router 192.168.30.1
dns-server 8.8.8.8
!
ip dhcp pool SERVERS
network 192.168.99.0 255.255.255.0
default-router 192.168.99.1
dns-server 8.8.8.8
```

Step 5: Basic Connectivity Test

- **Assign Static IPs Temporarily** (for initial testing):
 - Example for HR PC: IP 192.168.10.11, Subnet 255.255.255.0, Gateway 192.168.10.1.
 - Example for SALES PC: IP 192.168.30.11, Subnet 255.255.255.0, Gateway 192.168.30.1.

- Ping Test:
 - Ping within VLANs (e.g., HR PC1 to HR PC2, SALES PC1 to SALES PC2).
 - Resolve issues by checking
 - Correct VLAN assignments (`show vlan brief`).
 - Interface status (`show ip interface brief`), (`show interface status`)
 - Switch port modes (`show running-config`)

Step 6: DHCP Testing:

- Configure End Devices for DHCP:
 - On each PC, laptop, and server, go to Desktop > IP Configuration > DHCP.
- Verify IP Assignment:
 - On the router, use `show ip dhcp binding` to confirm IP assignments.
 - Verify IP, gateway, and DNS settings on each device.
- Inter-VLAN Ping Test:
 - Ping from HR PC (VLAN 10) to IT PC (VLAN 20).
 - Ping from SALES PC (VLAN 30) to a server in VLAN 99.
 - Ensure `ip routing` is enabled for inter-VLAN communication.

Step 7: Security and Optimization

Port Security: On all switches, secure access ports

```
interface range FastEthernet0/1 - 6
switchport port-security
switchport port-security maximum 2
switchport port-security violation shutdown
switchport port-security mac-address sticky
```

Key Concepts Implemented

- **VLAN Segmentation:** HR (VLAN 10), IT (VLAN 20), SALES (VLAN 30), and SERVERS (VLAN 99) are logically separated.
- **DHCP:** Dynamic IP assignment with reserved ranges for static devices.
- **Hybrid Interface Configuration:** Access ports for HR and IT, trunk port with subinterfaces for SALES and SERVERS.
- **Inter-VLAN Routing:** Enabled via router physical interfaces (HR, IT) and subinterfaces (SALES, SERVERS).
- **End-to-End Connectivity:** Verified within and across VLANs.

Troubleshooting Notes

- **APIPA (169.254.x.x) on PCs:**
 - Check DHCP service (`show ip dhcp server`).
 - Verify router interface status (`show ip interface brief`).
 - Ensure correct VLAN on switch ports (`show vlan brief`).
- **No Inter-VLAN Routing:**
 - Confirm `ip routing` is enabled.
 - Verify trunk port on SALES/SERVERS switch allows VLANs 30 and 99.
 - Check subinterface configurations (`show running-config`).
- **DHCP Issues:**
 - Confirm excluded address ranges.
 - Reboot devices (Config tab > Power Cycle) to refresh DHCP requests.
- **Connectivity Issues:**
 - Verify access port VLAN assignments.
 - Check trunk port allowed VLANs (`show interfaces trunk`).
 - Confirm routing table (`show ip route`).

Results

- All devices in HR, IT, SALES, and SERVERS received IPs via DHCP.
- Intra-VLAN communication (e.g., HR PC1 to HR PC2) and inter-VLAN communication (e.g., HR to SALES) confirmed via ping tests.

- SALES and SERVERS VLANs successfully separated on a single switch using a trunk link.

Conclusion

The enterprise network setup in Cisco Packet Tracer successfully achieved the objective of creating a segmented, scalable, and functional network for the HR, IT, SALES, and Server Room departments. By leveraging VLANs (10, 20, 30, and 99), the network was logically separated to enhance security and organization. The use of **access ports** for HR and IT switches and a **trunk port** for the combined SALES/SERVERS switch, paired with the ISR4331 router's physical interfaces (Gig0/0 for HR, Gig0/1 for IT) and subinterfaces (Gig0/2.30 for SALES, Gig0/2.99 for SERVERS), enabled efficient inter-VLAN routing. The DHCP configuration on the router dynamically assigned IPs to all devices, with reserved ranges to prevent conflicts, ensuring seamless IP allocation.

Testing confirmed intra-VLAN and inter-VLAN connectivity through ping tests. Troubleshooting steps addressed common issues like APIPA, VLAN misconfigurations, and interface status. Security measures, such as port security, were suggested to enhance network protection. The setup is robust for a small-to-medium enterprise, with clear pathways for scalability (e.g., NAT for internet access, HSRP for redundancy, or Layer 3 switching).